Octahedral complexes:

Octahedral complexes of the type $[MA_2B_4]^{n\pm}$, $[M(xx)_2B_2]^{n\pm}$ shows *cis-trans* isomerism. Here A and B are monodentate ligands and xx is bidentate ligand with two same kind of donor atoms. In the octahedral complex, the position of ligands is indicated by the following numbering scheme.

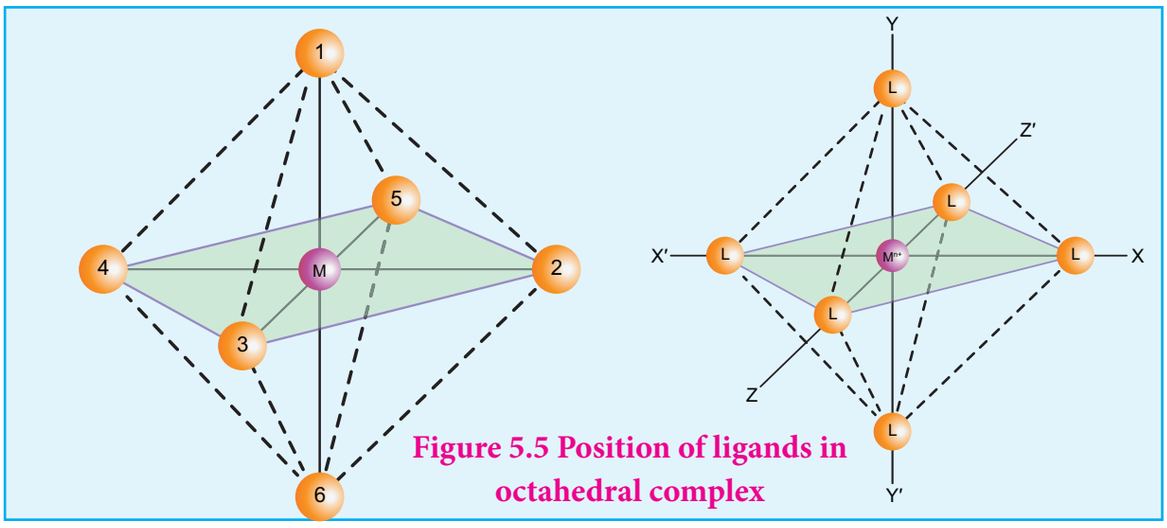


Figure 5.5 Position of ligands in octahedral complex

In the above scheme, the positions (1,2), (1,3), (1,4), (1,5), (2,3), (2,5), (2,6), (3,4), (3,6), (4,5), (4,6), and (5,6) are identical and if two similar groups are present in any one of these positions, the isomer is referred as a *cis* isomer. Similarly, positions (1,6), (2,4), and (3,5) are identical and if similar ligands are present in these positions it is referred as a *trans*-isomer.

Octahedral complex of the type $[MA_3B_3]^{n\pm}$ also shows geometrical isomerism. If the three similar ligands (A) are present in the corners of one triangular face of the octahedron and the other three ligands (B) are present in the opposing triangular face, then the isomer is referred as a facial isomer (*fac* isomer)- Figure 5.6 (a).

If the three similar ligands are present around the meridian which is an imaginary semicircle from one apex of the octahedral to the opposite apex as shown in the figure 5.6(b), the isomer is called as a meridional isomer (*mer* isomer). This is called meridional because each set of ligands can be regarded as lying on a meridian of an octahedron.

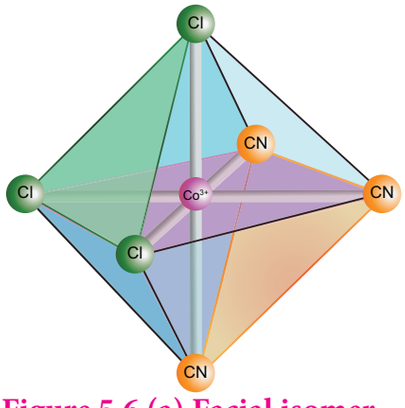


Figure 5.6 (a) Facial isomer

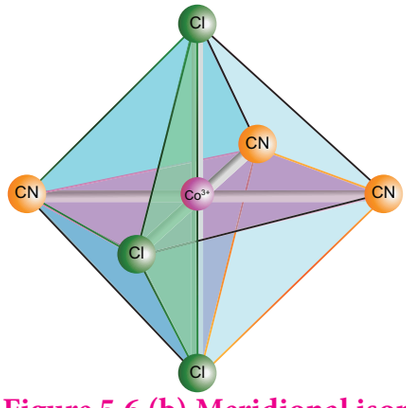


Figure 5.6 (b) Meridional isomer

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